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Manifold to selectively inflate seat airbags

Abstract

An assembly includes a seat. The assembly includes a first airbag supported by the seat. The assembly includes a second airbag supported by the seat. The assembly includes a manifold defining a chamber. The manifold includes a first fluid outlet operatively coupled to the first airbag to provide fluid from the chamber to the first airbag. The manifold includes a second fluid outlet operatively coupled to the second airbag to provide fluid from the chamber to the second airbag. The manifold includes an exhaust fluid outlet. The assembly includes an inflator operatively coupled to the manifold to provide fluid to the chamber.

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Background/Summary

BACKGROUND

(1) A vehicle may include one or more airbags deployable during certain vehicle impacts to control kinematics of occupants inside the vehicle. The airbag may be a component of an airbag assembly including a housing supporting the airbag, and an inflator in communication with the airbag for inflating the airbag from an uninflated position to an inflated position.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a perspective view of a portion of a vehicle broken away to show a passenger cabin with airbags at uninflated positions.
- (2) FIG. 2 is a perspective view of the portion of the vehicle broken away to show the passenger cabin the airbags at inflated positions.
- (3) FIG. 3 is a diagrammatic view of a manifold with valves at closed positions.
- (4) FIG. 4 is a diagrammatic view of the manifold with the valves at open positions.
- (5) FIG. 5 is a block diagram of components of the vehicle.
- (6) FIG. 6 is a flow chart showing a process for controlling inflation of the airbags.

DETAILED DESCRIPTION

- (7) An assembly includes a seat. The assembly includes a first airbag supported by the seat. The assembly includes a second airbag supported by the seat. The assembly includes a manifold defining a chamber. The manifold includes a first fluid outlet operatively coupled to the first airbag to provide fluid from the chamber the first airbag. The manifold includes a second fluid outlet operatively coupled to the second airbag to provide fluid from the chamber to the second airbag. The manifold includes an exhaust fluid outlet. The assembly includes an inflator operatively coupled to the manifold to provide fluid to the chamber.
- (8) The assembly may include an exhaust valve movable to an open position permitting fluid flow through the exhaust fluid outlet and to a closed position inhibiting fluid flow through the exhaust fluid outlet.
- (9) The assembly may include a first valve movable to an open position permitting fluid flow through the first fluid outlet and to a closed position inhibiting fluid flow through the first fluid outlet, and a second valve movable to an open position permitting fluid flow through the second fluid outlet and to a closed position inhibiting fluid flow through the second fluid outlet.
- (10) The assembly may include a computer having a processor and memory storing instructions executable by the processor to, in response to detecting a certain impact, command the first valve from a closed position to an open position and command actuation of the inflator.
- (11) The instructions may include instructions to command the first valve from the open position to

the closed position after commanding the first valve from the closed position to the open position.

(12) The instructions may include instructions to command the exhaust valve from a closed position to an open position concurrent with or after commanding the first valve from the open position to the closed position.

(13) The instructions may include instructions to determine whether an occupant of the seat is above a size threshold. The instructions may include instructions to, in response to determining the occupant of the seat is not above the size threshold, command the first valve from the open position to the closed position a first amount of time after commanding the first valve from the closed position to the open position. The instructions may include instructions to, in response to determining the occupant of the seat is above the size threshold, command the first valve from the open position to the closed position a second amount of time after commanding the first valve from the closed position to the open position, the second amount of time being greater than the first amount of time.

(14) The instructions may include instructions to command the exhaust valve from a closed position to an open position concurrent with, or after, commanding the first valve from the open position to the closed position.

(15) The assembly of claim may include a second seat, and the instructions may include instructions to, in response to both determining the second seat is occupied and detecting the certain impact, command the second valve from a closed position to an open position.

(16) The instructions may include instructions to command the second valve from the closed position to the open position a threshold amount of time after commanding the first valve from the closed position to the open position.

(17) The threshold amount of time may be 20-50 milliseconds.

(18) The instructions may include instructions to command the first valve from the open position to the closed position after commanding the first valve from the closed position to the open position, and to command the second valve from the open position to the closed position after commanding the second valve from the closed position to the open position.

(19) The instructions may include instructions to command the exhaust valve from a closed position to an open position concurrent with or after commanding the second valve from the open position to the closed position.

(20) The instructions may include instructions to determine whether an occupant of the seat is above a size threshold. The instructions may include instructions to, in response to determining the occupant of the seat is not above the size threshold, command the first valve from the open position to the closed position a first amount of time after commanding the first valve from the closed position to the open position. The instructions may include instructions to, in response to determining the occupant of the seat is above the size threshold, command the first valve from the open position to the closed position a second amount of time after commanding the first valve from the closed position to the open position, the second amount of time being greater than the first amount of time. The instructions may include instructions to command the second valve from the open position to the closed position.

(21) The instructions may include instructions to command the exhaust valve from a closed position to an open position concurrent with or after commanding the second valve from the open position to the closed position.

(22) The chamber may define a first cross-sectional area and the first fluid outlet may define a second cross-sectional area that is less than the first cross-sectional area.

(23) The manifold and the inflator may be supported by the seat.

(24) With reference to the Figures, wherein like numerals indicate like parts throughout the several views, an assembly **10** for controlling kinematics of an occupant of a vehicle **12** is shown. The assembly **10** includes a seat **14**. The assembly **10** includes a first airbag **16** supported by the seat **14**. The assembly **10** includes a second airbag **18** supported by the seat **14**. The assembly **10** includes a

manifold **20** defining a chamber **22**. The manifold **20** includes a first fluid outlet **24** operatively coupled to the first airbag **16** to provide fluid from the chamber **22** to the first airbag **16**. The manifold **20** includes a second fluid outlet **26** operatively coupled to the second airbag **18** to provide fluid from the chamber **22** to the second airbag **18**. The manifold **20** includes an exhaust fluid outlet **28**. The assembly **10** includes an inflator **30** operatively coupled to the manifold **20** to provide fluid to the chamber **22**. The manifold **20** may temporarily store inflation medium for inflating the first airbag **16** and the second airbag **18** to provide increased control of inflation dynamics of the first airbag **16** and the second airbag **18**, e.g., control of inflation timing and airbag inflation pressure. The exhaust fluid outlet **28** permits excess inflation medium to be purged from the manifold **20**, e.g., after inflation of the first airbag **16** and/or the second airbag **18**.

(25) With reference to FIGS. **1** and **2**, the vehicle **12** may be any suitable type of automobile, e.g., a passenger or commercial automobile such as a sedan, a coupe, a truck, a sport utility vehicle, a crossover vehicle, a van, a minivan, a taxi, a bus, etc.

(26) In the present description, relative vehicular orientations and directions (by way of example, top, bottom, front, rear, outboard, inboard, inward, outward, forward, rearward, lateral, left, right, etc.) are from the perspective of an occupant seated in the vehicle **12** and facing forward, e.g., toward a forward windshield of the vehicle **12**. The forward direction of the vehicle **12** is the direction of movement of the vehicle **12** when the vehicle **12** is engaged in forward drive with wheels of the vehicle straight.

(27) The vehicle **12** defines a lateral axis **A1** extending between a left-side and a right-side of the vehicle **12**. The vehicle **12** defines a longitudinal axis **A2** extending between a front and a rear of the vehicle **12**. The vehicle **12** defines a vertical axis extending between a top and a bottom of the vehicle **12**. The lateral axis **A1**, the longitudinal axis **A2**, and the vertical axis are perpendicular relative to each other.

(28) The vehicle **12** includes a frame and a body. The body may be of unitary construction, in which the frame is unitary with the body, e.g., including frame rails, rockers, pillars, roof rails, etc. As another example, the body and frame may have a body-on-frame construction (also referred to as a cab-on-frame construction) in which the body (including rockers, pillars, roof rails, etc.) and frame are separate components, i.e., are modular, and the body is supported on and affixed to the frame. Alternatively, the frame and body may have any suitable construction. The frame and the body may be of any suitable material, for example, steel, aluminum, and/or fiber-reinforced plastic, etc. The body defines the passenger cabin to house occupants, if any, of the vehicle **12**. The passenger cabin may extend across the vehicle **12**, i.e., from one side to the other side of the vehicle **12**. The passenger cabin includes the front end and a rear end.

(29) One or more seats **14**, **32**, e.g., the seat **14** and a second seat **32**, may be supported in the passenger cabin, e.g., by the floor of the vehicle **12**. The seat **14** and the second seat **32** may be spaced arranged along the lateral axis **A1**, e.g., as right and left seats **14**, **32**. Each seat **14** includes a seatback **34** and a seat bottom **36** that can support an occupant of the seat **14**. For example, the occupant of the seat **14** may sit atop a top surface of the seat bottom **36** and recline against the seatback **34**. The seatback **34** and the seat bottom **36** may be adjustable in multiple degrees of freedom. Specifically, the seatback **34** and the seat bottom **36** may themselves be adjustable, in other words, adjustable components within the seatback **34** and/or the seat bottom **36**.

(30) The seatback **34** can include a frame and a covering. The frame may include tubes, beams, etc. Specifically, the frame may include a pair of upright frame members. The upright frame members are elongated, and specifically, are elongated in a generally upright direction (e.g., generally along a vertical axis) when the seatback **34** is in a generally upright position. The upright frame members are spaced from each other along the lateral axis **A1**. The frame may include one or more cross-members extending between the upright frame members. The frame, including the upright frame members, may be of any suitable plastic material, e.g., carbon fiber reinforced plastic (CFRP), glass fiber-reinforced semi-finished thermoplastic composite (organosheet), etc. As another

example, some or all components of the frame may be formed of a suitable metal, e.g., steel, aluminum, etc. The cover may include upholstery and padding. The upholstery may be cloth, leather, faux leather, or any other suitable material. The upholstery may be stitched in panels around the frame. The padding may be between the cover and the frame and may be foam or any other suitable material.

(31) The seat bottom **36** can include a frame and a covering. The frame may include tubes, beams, etc. Specifically, the frame may include a pair of frame members, e.g., elongated along the longitudinal axis **A2**. The frame members are spaced from each other along the lateral axis **A1**. The frame may include cross-members extending between the frame members. The cross-members may be elongated along the lateral axis **A1**. The frame can include a seat pan. The seat pan may be generally planar and extend from one of the frame members to the other of the frame members, e.g., along the lateral axis **A1**. The seat pan may be fixed to the frame members. The frame, including the frame members, the cross-members, and the seat pan may be of any suitable plastic material, e.g., carbon fiber reinforced plastic (CFRP), glass fiber-reinforced semi-finished thermoplastic composite (organosheet), etc. As another example, some or all components of the frame may be formed of a suitable metal, e.g., steel, aluminum, etc. The covering may include upholstery and padding. The upholstery may be cloth, leather, faux leather, or any other suitable material. The upholstery may be stitched in panels around the frame. The padding may be between the upholstery and the frame. The padding may be foam or any other suitable material. The seatback **34** is supported by the seat bottom **36** at a rear of the seat bottom **36**. The seat bottom **36** extends forward from the seatback **34**. The seatback **34** is movable relative to the seat bottom **36**. For example, the seatback **34** may pivot about an axis that is parallel to the longitudinal axis **A2**.

(32) The vehicle **12** includes the first airbag **16** and the second airbag **18** for controlling kinematics of an occupant of the vehicle **12**. Each of the airbags **16**, **18** is inflatable from an uninflated position, shown in FIG. 1, to an inflated position, shown in FIG. 2. The airbags **16**, **18** may each include panels of a woven polymer, or any other suitable material, that define inflation chamber **22s** that can receive inflation medium to inflate the first airbag **16** and the second airbag **18** to the inflated positions. As one example, the first airbag **16** and the second airbag **18** may be formed of woven nylon yarn, for example, nylon 6-6. Other examples include polyether ether ketone (PEEK), polyetherketoneketone (PEKK), polyester, etc. The woven polymer may include a coating, such as silicone, neoprene, urethane, etc. For example, the coating may be polyorgano siloxane.

(33) The first airbag **16** and the second airbag **18** are supported by the seat **14**. The first airbag **16** and the second airbag **18** may be supported by the seatback **34** of the seat **14**. The first airbag **16** and the second airbag **18** may be fixed to frame of the seatback **34**, e.g., fixed to the pair of upright frame members via fasteners or other suitable structure. The first airbag **16** and the second airbag **18** may be supported via airbag housings, e.g., fixed to the frame of the seatback **34**. The first airbag **16** and the second airbag **18** in the uninflated positions may be internal of the seat **14**, e.g., under the covering of the seatback **34**. The second airbag **18** may be vehicle-inboard of the first airbag **16**. The second airbag **18** may be between the first airbag **16** and the second seat **32** along the lateral axis.

(34) The inflator **30** provides inflation medium, e.g., to inflate the first airbag **16** and/or the second airbag **18** from the uninflated positions to the inflated positions. The inflator **30** may be, for example, a pyrotechnic inflator that uses a chemical reaction to drive inflation medium to the first airbag **16** and/or the second airbag **18**. The inflator **30** may be of any suitable type, for example, a cold-gas inflator.

(35) The inflator **30** is supported by the seat **14**. The inflator **30** may be supported by the seatback **34** of the seat **14**. The inflator **30** may be fixed to frame of the seatback **34**, e.g., fixed to the pair of upright frame members via fasteners or other suitable structure. The inflator **30** may be fixed to another component supported by the seat **14**, e.g., to the manifold **20**.

(36) The inflator **30** is operatively coupled to the manifold **20** to provide fluid, such as inflation

medium, to the chamber **22** of the manifold **20**. For example, the inflator **30** may be fixed to the manifold **20** and provide inflation medium directly to the chamber **22**. As another example, the inflator **30** may be connected to the manifold **20** via fill tube or other suitable structure that directs inflation medium from the inflator **30** to the chamber **22** of the manifold **20**.

(37) The vehicle **12** may include additional airbags (not shown). For example, the vehicle **12** may include one or more airbags supported by the second seat **32**, by an instrument panel of the vehicle **12**, by a steering wheel of the vehicle **12**, by a pillar of the body of the vehicle **12**, or any other suitable structure of the vehicle **12**. The vehicle **12** may include additional manifolds, inflators, etc., to provide inflation medium to the additional airbags (not shown), e.g., supported by the second seat **32** and as described herein.

(38) The manifold **20** is included in the assembly **10** to receive inflation medium from the inflator **30** and to provide the inflation medium to the first airbag **16** and/or the second airbag **18**. The manifold **20** is supported by the seat **14**. The manifold **20** may be supported by the seatback **34** of the seat **14**. The manifold **20** may be fixed to frame of the seatback **34**, e.g., fixed to the pair of upright frame members via fasteners or other suitable structure. The manifold **20** may be metal, a fiber laminate, or any other suitable material.

(39) The manifold **20** defines the chamber **22** to receive and provide the inflation medium. The manifold **20** encloses the chamber **22**. The chamber **22** may store inflation medium from the inflator **30** under pressure. The chamber **22** may be of sufficient volume such that generally inflation medium under constant pressure can be provided to the first airbag **16** and/or second airbag **18**. The manifold **20** may be a pressure vessel that stores a sufficient volume of fluid at sufficient pressure to fully inflate the first airbag **16** and the second airbag **18** to the inflated positions. The volume of the chamber **22** may be based on a sum of the volumes of the airbags **16**, **18** in the inflated positions and an amount of pressure that is sufficient to cause inflation of the airbags **16**, **18** within a certain amount of time. The chamber **22** defines a cross-sectional area that is greater than a cross-sectional area defined by the first fluid outlet **24**. In other words, the cross-sectional area of the first fluid outlet **24** is less than the cross-sectional area of the chamber **22**. The cross-sectional area of the first fluid outlet **24** extends perpendicular to a flow direction of fluid through the first fluid outlet **24**. The cross-sectional area of the chamber **22** extends perpendicular to a flow direction of fluid through the chamber **22**.

(40) The manifold **20** includes a plurality of fluid outlets **24**, **26**, **28**, e.g., the first fluid outlet **24**, the second fluid outlet **26**, and an exhaust fluid outlet **28**, for directing fluid flow out of the chamber **22**. The fluid outlets **24**, **26**, **28** provide passages that direct the flow of fluid. The fluid outlets **24**, **26**, **28** may be open to the chamber **22** on one end and open to external of the chamber **22** on an opposite end. The first fluid outlet **24** may be operatively coupled to the first airbag **16** to provide fluid from the chamber **22** to the first airbag **16**. The second fluid outlet **26** may be operatively coupled to the second airbag **18** to provide fluid from the chamber **22** to the second airbag **18**. For example, a fill tube **38** may connect the first fluid outlet **24** to the first airbag **16** and may connect the second fluid outlet **26** to the second airbag **18**. The exhaust fluid outlet **28** enables fluid in the chamber **22** to be depressurized, e.g., the pressurized fluid may be exhausted from the chamber **22** without such fluid being directed to any airbag. For example, one end of the exhaust fluid outlet **28** may be open to the chamber **22** and an opposite end may be open to external of the chamber **22** and not operatively coupled to any airbag. The exhaust fluid outlet **28** may exhaust fluid, e.g., generally into to the passenger cabin.

(41) With reference to FIGS. **3** and **4**, one or more valves, e.g., a first valve **40**, a second valve **42**, and an exhaust valve **44**, may be included for selectively permitting and inhibiting fluid flow through the fluid outlets **24**, **26**, **28**. Each valve **40**, **42**, **44** is movable to a closed position that inhibits fluid flow, e.g., as shown in FIG. **3**. Each valve **40**, **42**, **44** is movable to an open position that permits fluid flow, e.g., as shown in FIG. **4**. The valves **40**, **42**, **44** may be, for example, ball valves, butterfly valves, gate valves, or other suitable valve structure that selectively permits and

inhibits fluid flow, including in some examples those that are conventionally known. The valves **40, 42, 44** may move from the closed position to the open position, or vice versa, in response to a command from a computer **46**. Each valve **40, 42, 44** may include an electromagnetic solenoid, a motor, and/or other suitable structure, including in some examples those that are conventionally known, for moving the respective valve from the closed position to the open position, or vice versa, in response to a command from the computer **46**.

(42) The first valve **40** may be connected at the first fluid outlet **24**. The first valve **40** at the open position may permit fluid flow through the first fluid outlet **24** from the chamber **22** to the fill tube **38** operatively coupling the first fluid outlet **24** to the first airbag **16**. The first valve **40** at the closed position may inhibit fluid flow through the first fluid outlet **24** from the chamber **22** to the fill tube **38** operatively coupling the first fluid outlet **24** to the first airbag **16**.

(43) The second valve **42** may be connected at the second fluid outlet **26**. The second valve **42** at the open position may permit fluid flow through the second fluid outlet **26** from the chamber **22** to the fill tube **38** operatively coupling the second fluid outlet **26** to the second airbag **18**. The second valve **42** at the closed position may inhibit fluid flow through the second fluid outlet **26** from the chamber **22** to the fill tube **38** operatively coupling the second fluid outlet **26** to the second airbag **18**.

(44) The exhaust valve **44** may be connected at the exhaust fluid outlet **28**. The exhaust valve **44** at the open position may permit fluid flow out of the chamber **22** through the exhaust fluid outlet **28**. The exhaust valve **44** at the closed position may inhibit fluid flow out of the chamber **22** through the exhaust fluid outlet **28**.

(45) With reference to FIG. 5, the vehicle **12** may include at least one impact sensor **48**. The impact sensor **48** is configured to detect certain vehicle impacts. A “certain vehicle impact” is an impact of the type and/or magnitude for which inflation of the first airbag **16** is designed i.e., “certain” indicates the type and/or magnitude of the impact. The type and/or magnitude of such “certain vehicle impacts” may be pre-stored in the computer **46**, e.g., in a restraints control module and/or a body control module. The impact sensor **48** may be configured to detect a certain impact contemporaneously with the certain impact and/or the impact sensor **48** may be configured to sense certain impacts prior to impact, i.e., pre-impact sensing. The impact sensor **48** may be of any suitable type, for example, post contact sensors such as accelerometers, pressure sensors, and contact switches; and pre-impact sensors such as radar, LIDAR, and vision sensing systems. The vision systems may include one or more cameras, CCD image sensors, CMOS image sensors, etc. The impact sensors **48** may be located at numerous points in or on the vehicle **12**.

(46) The vehicle **12** may include one or more occupancy sensors **50** that detect occupants of the seats **14, 32**. The occupancy sensors **50** may detect whether or not the seat **14** and/or the second seat **32** are occupied, i.e., whether occupants are seated in the seat **14** and/or the second seat **32**. The occupancy sensors **50** may detect a size of the occupant of the seat **14** and/or the second seat **32**, e.g., a weight, a height, etc. The occupancy sensors **50** may include, for example, visible-light or infrared cameras directed at the seat **14**, weight sensors supported by the seat bottom **36**, or other suitable structure, including in some examples those that are conventionally known.

(47) The vehicle **12** may include a communication network **52**. The communication network **52** includes hardware, such as a communication bus, for facilitating communication among components of the vehicle **12**, e.g., the inflator **30**, the valves **40, 42, 44**, the computer **46**, the impact sensors **48**, the occupancy sensors **50**, etc. The communication network **52** may facilitate wired or wireless communication among the components in accordance with a number of communication protocols such as controller area network (CAN), Ethernet, WiFi, Local Interconnect Network (LIN), and/or other wired or wireless mechanisms. Alternatively or additionally, in cases where the computer **46** comprises a plurality of devices, the communication network **52** may be used for communications between devices represented as the computer **46** in this disclosure.

(48) The computer **46** may be, for example, a restraints control module of the vehicle **12**. The computer **46** includes a processor, a memory, etc. The memory of the computer **46** may include memory for storing programming instructions executable by the processor as well as for electronically storing data and/or databases. The computer **46** may be a microprocessor-based computer implemented via circuits, chips, or other electronic components. For example, the computer **46** can be a generic computer **46** with a processor and memory as described above and/or may include an electronic control unit (ECU) or controller for a specific function or set of functions, and/or a dedicated electronic circuit including an ASIC that is manufactured for a particular operation, e.g., an ASIC for processing sensor data and/or communicating the sensor data. In another example, computer **46** may include an FPGA (Field-Programmable Gate Array) which is an integrated circuit manufactured to be configurable by a user. Typically, a hardware description language such as VHDL (Very High-Speed Integrated Circuit Hardware Description Language) is used in electronic design automation to describe digital and mixed-signal systems such as FPGA and ASIC. For example, an ASIC is manufactured on VHDL programming provided pre-manufacturing, whereas logical components inside an FPGA may be configured based on VHDL programming, e.g., stored in a memory electrically connected to the FPGA circuit. In some examples, a combination of processor(s), ASIC(s), and/or FPGA circuits may be included in the computer **46**. The memory can be of any type, e.g., hard disk drives, solid state drives, servers, or any volatile or non-volatile media. The memory can store the collected data sent from the sensors.

(49) The computer **46** is programmed to, i.e., the memory stores instructions executable by the processor to, detect one or more certain impacts to the vehicle **12**. The computer **46** may detect one or more certain impacts based on information received, e.g., via the communication network **52**, from the impact sensors **48**. The information received from the impact sensors **48** may indicate, e.g., a location of impact, angle of impact, amount of force of impact, acceleration of the vehicle **12**, etc.

(50) The computer **46** is programmed to determine whether the seat **14** and/or the second seat **32** are occupied. The computer **46** may determine whether the seat **14** and/or the second seat **32** are occupied based on information received, e.g., via the communication network **52**, from one or more of the occupancy sensors **50**. The information received from the occupancy sensors **50** may indicate, e.g., the seat **14** is occupied and/or that the second seat **32** is occupied.

(51) The computer **46** is programmed to determine whether an occupant of the seat **14** is above a size threshold. The computer **46** may determine whether the occupant of the seat **14** is above the size threshold by comparing an identified size of the occupant with the size threshold. The size of the occupant may be, e.g., a weight, a height, etc., of the occupant. The computer **46** may identify the size of the occupant based on information received, e.g., via the communication network **52**, from one or more of the occupancy sensors **50**. The information received from the occupancy sensors **50** may indicate, e.g., a weight of the occupant of the seat **14**, a height of the occupant of the seat **14**, etc. The threshold size may be empirically determined and/or calculated, e.g., based on crash test data, computer modeling, etc.

(52) The computer **46** may be programmed to command actuation of the inflator **30** to provide inflation medium to the manifold **20**. The computer **46** may command actuation of the inflator **30** by transmitting a command to the inflator **30**, e.g., via the communication network **52**. The command may include, for example, an electrical pulse that triggers expansion of the inflation medium.

(53) The computer **46** may be programmed to command one or more of the valves **40**, **42**, **44** from the closed position to the open position, and vice versa. The computer **46** may command one or more of the valves **40**, **42**, **44** to the open position or the closed position by transmitting one or more commands specifying the open position or the closed position to one or more of the valves **40**, **42**, **44**. For example, the computer **46** may command the first valve **40** from the closed position to the open position by transmitting a command to the first valve **40**, the command specifying the

open position. As another example, the computer **46** may command the exhaust valve **44** from the open position to the closed position by transmitting a command to the exhaust valve **44**, the command specifying the closed position.

(54) The computer **46** may actuate the inflator **30** and the valves **40**, **42**, **44** to control inflation of the first airbag **16** and the second airbag **18**, e.g., in response to detecting a certain impact to the vehicle **12**. For example, the computer **46** may actuate the inflator **30** and actuate the first valve **40** and/or the second valve **42** to the open position to cause inflation medium to flow to the first airbag **16** and/or the second airbag **18**.

(55) The computer **46** may actuate the second valve **42** to the open position in response to determining the second seat **32** is occupied, e.g., to permit the second airbag **18** to control kinematics of the occupant of the second seat **32**. The computer **46** may actuate one of the first valve **40** or the second valve **42** to the open position after the other to inflate one airbag **16**, **18** after the other airbag **16**, **18**. Inflating one airbag **16**, **18** after the other airbag **16**, **18** may control kinematics of the occupants of the first seat **14** and the second seat **32** differently than inflating both airbags **16**, **18** at a same time. The computer **46** may actuate one of the first valve **40** or the second valve **42** to the open position a threshold amount of time after the other to inflate one airbags **16**, **18** after the other. The threshold amount of time may be empirically determined, and/or calculated, e.g., based on crash test data, computer modeling, etc. The threshold amount of time is greater than zero and may be, e.g., 20-50 milliseconds.

(56) The computer **46** may be programmed command the first valve **40** and/or the second valve **42** to the closed position a certain amount time after actuating them to the open position, e.g., to control an inflation pressure of the respective airbag in the inflated position. A longer amount of time between opening and then closing the valves **40**, **42** permits more inflation medium to flow into the airbag, thereby providing a relatively higher inflation pressure to the respective airbag. A shorter amount of time between opening and then closing the valves **40**, **42** permits less inflation medium to flow into the airbag, thereby providing a lower inflation pressure to the respective airbag.

(57) The computer **46** may control the timing of opening and then closing one or more of the valves **40**, **42** based on a detected size of the occupant of the seat **14**, e.g., by comparing the detected size of the occupant with the size threshold. In response to determining that the occupant of the seat **14** is not above the size threshold, the computer **46** may command the first valve **40** from the open position to the closed position a first amount of time after commanding the first valve **40** from the closed position to the open position. In response to determining that the occupant of the seat **14** is above the size threshold, the computer **46** may command the first valve **40** from the open position to the closed position a second amount of time after commanding the first valve **40** from the closed position to the open position. The first amount of time may be less than the second amount of time and provide a lower airbag pressure for the occupant that is not above the size threshold. The second amount of time may be greater than the first amount of time and provide a higher airbag pressure for the occupant that is above the size threshold. The first amount of time and the second amount of may be empirically determined, and/or calculated, e.g., based on crash test data, computer modeling, etc.

(58) The computer **46** may be programmed to command the exhaust valve **44** from the closed position to the open position, e.g., to release pressurized inflation medium from the manifold **20**. The computer **46** may command the exhaust valve **44** to the open position concurrent with or after commanding the first valve **40** and/or the second valve **42** from the open position to the closed position. For example, the computer **46** may wait to command the exhaust valve **44** to the open position until after the first airbag **16** and the second airbag **18** are inflated and after both the first valve **40** and the second valve **42** are commanded to the closed position.

(59) FIG. **6** is a process flow diagram illustrating an exemplary process **600** for controlling the inflation of the first airbag **16** and the second airbag **18**. The process **600** may be performed by the

computer **46**. In other words, the memory of the computer **46** may store instructions executable by the processor to perform the process **600**. The process **600** begins in a block **605**, in which the computer **46** collects data, e.g., received from the impact sensor **48**, the occupancy sensors **50**, etc., e.g., via the communication network **52**. The computer **46** may continue to receive data throughout the process **600**. Throughout the process **600** in the present context means substantially continuously or at time intervals, e.g., every 5 milliseconds. The computer **46** may store the collected data, as well as other determinations, detections, commands, etc., received, performed, or transmitted throughout the process **600**. The valves **40**, **42**, **44** may all initially be at the closed position when the process **600** begins at the block **605**.

(60) At block **610**, the computer **46** determines whether a certain impact to the vehicle **12** has been detected, e.g., based on information received from the impact sensor **48** via the communication network **52** and as described above. Upon determining the certain impact has been detected, the computer **46** moves to a block **615**. Upon determining the certain impact has not been detected, the computer **46** returns to the block **605**.

(61) At the block **615**, the computer **46** determines whether the second seat **32** is occupied, e.g., based on information received from the occupancy sensors **50** via the communication network **52** and as described herein. Upon determining that the second seat **32** is not occupied, the computer **46** moves to a block **620**. Upon determining that the second seat **32** is occupied, the computer **46** moves to a block **645**.

(62) At the block **620** and in response to detecting the certain impact to the vehicle **12**, the computer **46** commands actuation of the inflator **30** and commands the first valve **40** from the closed position to the open position to inflate the first airbag **16**, e.g., at substantially a same time or one after the other and as described above.

(63) At a block **625**, the computer **46** determines whether the occupant of the first seat **14** is above the threshold size, e.g., based on information received from the occupancy sensors **50** via the communication network **52** and as described herein. Upon determining the occupant is not above the threshold size the computer **46** moves to a block **630**. Upon determining the occupant is above the threshold size the computer **46** moves to a block **635**. The computer **46** may execute the block **625** at substantially a same time as the block **620**.

(64) At the block **630** and in response to determining the occupant of the seat **14** is not above the size threshold, the computer **46** commands the first valve **40** from the open position to the closed position a first amount of time after commanding the first valve **40** from the closed position to the open position at the block **620**, e.g., as described above. After the block **630** the computer **46** may move to a block **640**.

(65) At the block **635** and in response to determining the occupant of the seat **14** is above the size threshold, the computer **46** commands the first valve **40** from the open position to the closed position a second amount of time after commanding the first valve **40** from the closed position to the open position at the block **620**, e.g., as described above. The second amount of time is greater than the first amount of time, e.g., to provide increased inflation pressure to the first airbag **16** to control kinematics of the occupant that is above the threshold size. After the block **635** the computer **46** may move to the block **640**.

(66) At the block **640** the computer **46** commands the exhaust valve **44** from the closed position to the open position to purge any inflation medium remaining in the manifold **20**, e.g., as described above. The computer **46** may move the exhaust valve **44** to the open position concurrent with or after commanding the first valve **40** from the open position to the closed position at the block **630** or the block **635**. After the block **640** the process **600** may end.

(67) At the block **645** and in response to both determining the second seat **32** is occupied and detecting the certain impact to the vehicle **12**, the computer **46** commands the first valve **40** from the closed position to the open position, commands the second valve **42** from the closed position to the open position and commands actuation of the inflator **30** to inflate the first airbag **16** and the

second airbag **18**, e.g., as described above. The computer **46** may actuate the inflator **30** and command the first valve **40** to the open position at substantially a same time or one after the other. The computer **46** may command the second valve **42** from the closed position to the open position the threshold amount of time, e.g., 20-50 milliseconds, after commanding the first valve **40** to the open position.

(68) At a block **650**, the computer **46** determines whether the occupant of the first seat **14** is above the threshold size, e.g., based on information received from the occupancy sensors **50** via the communication network **52** and as described herein. Upon determining the occupant is not above the threshold size the computer **46** moves to a block **655**. Upon determining the occupant is above the threshold size the computer **46** moves to a block **660**. The computer **46** may execute the block **650** at substantially a same time as the block **645**.

(69) At the block **655** and in response to determining the occupant of the seat **14** is not above the size threshold, the computer **46** commands the first valve **40** from the open position to the closed position the first amount of time after commanding the first valve **40** from the closed position to the open position at the block **645**, e.g., as described above. Additionally, the computer **46** commands the second valve **42** to the closed position. The computer **46** may command the second valve **42** to the closed position at a certain amount of time after commanding the second valve **42** to the open position, e.g., based on a detected size of the occupant of the second seat **32**. After the block **655** the computer **46** may move to a block **665**.

(70) At the block **660** and in response to determining the occupant of the seat **14** is above the size threshold, the computer **46** commands the first valve **40** from the open position to the closed position the second amount of time after commanding the first valve **40** from the closed position to the open position at the block **645**, e.g., as described above. Additionally, the computer **46** commands the second valve **42** to the closed position, also as described at the block **655**. After the block **660** the computer **46** may move to a block **665**.

(71) At the block **665** the computer **46** commands the exhaust valve **44** from the closed position to the open position to purge and inflation medium remaining in the manifold **20**, e.g., as described above. The computer **46** may move the exhaust valve **44** to the open position concurrent with or after the latter of commanding the first valve **40** to the closed position or commanding the second valve **42** to the closed position the at the block **655** or the block **660**. After the block **640** the process **600** may end.

(72) In general, the computing systems and/or devices described may employ any of a number of computer operating systems, including, but by no means limited to, versions and/or varieties of the Ford Sync® application, AppLink/Smart Device Link middleware, the Microsoft Automotive® operating system, the Microsoft Windows® operating system, the Unix operating system (e.g., the Solaris® operating system distributed by Oracle Corporation of Redwood Shores, California), the AIX UNIX operating system distributed by International Business Machines of Armonk, New York, the Linux operating system, the Mac OSX and iOS operating systems distributed by Apple Inc. of Cupertino, California, the BlackBerry OS distributed by Blackberry, Ltd. of Waterloo, Canada, and the Android operating system developed by Google, Inc. and the Open Handset Alliance, or the QNX® CAR Platform for Infotainment offered by QNX Software Systems. Examples of computing devices include, without limitation, an on-board vehicle computer, a computer workstation, a server, a desktop, notebook, laptop, or handheld computer, or some other computing system and/or device.

(73) Computing devices generally include computer-executable instructions, where the instructions may be executable by one or more computing devices such as those listed above. Computer executable instructions may be compiled or interpreted from computer programs created using a variety of programming languages and/or technologies, including, without limitation, and either alone or in combination, Java™, C, C++, Matlab, Simulink, Stateflow, Visual Basic, Java Script, Python, Perl, HTML, etc. Some of these applications may be compiled and executed on a virtual

machine, such as the Java Virtual Machine, the Dalvik virtual machine, or the like. In general, a processor (e.g., a microprocessor) receives instructions, e.g., from a memory, a computer readable medium, etc., and executes these instructions, thereby performing one or more processes, including one or more of the processes described herein. Such instructions and other data may be stored and transmitted using a variety of computer readable media. A file in a computing device is generally a collection of data stored on a computer readable medium, such as a storage medium, a random access memory, etc.

(74) A computer-readable medium (also referred to as a processor-readable medium) includes any non-transitory (e.g., tangible) medium that participates in providing data (e.g., instructions) that may be read by a computer (e.g., by a processor of a computer). Such a medium may take many forms, including, but not limited to, non-volatile media and volatile media. Instructions may be transmitted by one or more transmission media, including fiber optics, wires, wireless communication, including the internals that comprise a system bus coupled to a processor of a computer. Common forms of computer-readable media include, for example, RAM, a PROM, an EPROM, a FLASH-EEPROM, any other memory chip or cartridge, or any other medium from which a computer can read.

(75) Databases, data repositories or other data stores described herein may include various kinds of mechanisms for storing, accessing, and retrieving various kinds of data, including a hierarchical database, a set of files in a file system, an application database in a proprietary format, a relational database management system (RDBMS), a nonrelational database (NoSQL), a graph database (GDB), etc. Each such data store is generally included within a computing device employing a computer operating system such as one of those mentioned above, and are accessed via a network in any one or more of a variety of manners. A file system may be accessible from a computer operating system, and may include files stored in various formats. An RDBMS generally employs the Structured Query Language (SQL) in addition to a language for creating, storing, editing, and executing stored procedures, such as the PL/SQL language mentioned above.

(76) In some examples, system elements may be implemented as computer-readable instructions (e.g., software) on one or more computing devices (e.g., servers, personal computers, etc.), stored on computer readable media associated therewith (e.g., disks, memories, etc.). A computer program product may comprise such instructions stored on computer readable media for carrying out the functions described herein.

(77) With regard to the media, processes, systems, methods, heuristics, etc. described herein, it should be understood that, although the steps of such processes, etc. have been described as occurring according to a certain ordered sequence, such processes could be practiced with the described steps performed in an order other than the order described herein. It further should be understood that certain steps could be performed simultaneously, that other steps could be added, or that certain steps described herein could be omitted. Operations, systems, and methods described herein should always be implemented and/or performed in accordance with an applicable owner's/user's manual and/or safety guidelines.

(78) The numerical adjectives “first,” “second,” etc., are used herein merely as identifiers and do not signify order or importance. Use of “in response to,” “based on,” and “upon” herein indicates a causal relationship, not merely a temporal relationship.

(79) The disclosure has been described in an illustrative manner, and it is to be understood that the terminology which has been used is intended to be in the nature of words of description rather than of limitation. Many modifications and variations of the present disclosure are possible in light of the above teachings, and the disclosure may be practiced otherwise than as specifically described.

Claims

1. An assembly, comprising: a seat; a first airbag supported by the seat; a second airbag supported by the seat; a manifold defining a chamber and including a first fluid outlet operatively coupled to the first airbag to provide fluid from the chamber to the first airbag, a second fluid outlet operatively coupled to the second airbag to provide fluid from the chamber to the second airbag, and an exhaust fluid outlet; the exhaust fluid outlet being decoupled from the first airbag, the exhaust fluid outlet having one end open to the chamber and another end open to atmosphere around the seat; and an inflator operatively coupled to the manifold to provide fluid to the chamber.
2. The assembly of claim 1, further comprising an exhaust valve moveable to an open position permitting fluid flow through the exhaust fluid outlet and to a closed position inhibiting fluid flow through the exhaust fluid outlet.
3. The assembly of claim 1, further comprising an exhaust valve at the exhaust fluid outlet.
4. An assembly, comprising: a seat; a first airbag supported by the seat; a second airbag supported by the seat; a manifold defining a chamber and including a first fluid outlet operatively coupled to the first airbag to provide fluid from the chamber to the first airbag, a second fluid outlet operatively coupled to the second airbag to provide fluid from the chamber to the second airbag, and an exhaust fluid outlet; an inflator operatively coupled to the manifold to provide fluid to the chamber; and an exhaust valve moveable to an open position permitting fluid flow through the exhaust fluid outlet and to a closed position inhibiting fluid flow through the exhaust fluid outlet, the exhaust fluid outlet having one end open to the chamber and another end open to atmosphere around the seat.
5. The assembly of claim 4, further comprising a first valve movable to an open position permitting fluid flow through the first fluid outlet and to a closed position inhibiting fluid flow through the first fluid outlet, and a second valve movable to an open position permitting fluid flow through the second fluid outlet and to a closed position inhibiting fluid flow through the second fluid outlet.
6. The assembly of claim 5, further comprising a computer having a processor and memory storing instructions executable by the processor to, in response to detecting a certain impact, command the first valve from a closed position to an open position and command actuation of the inflator.
7. The assembly of claim 6, wherein the instructions include instructions to command the first valve from the open position to the closed position after commanding the first valve from the closed position to the open position.
8. The assembly of claim 7, wherein the instructions include instructions to command the exhaust valve from the closed position to the open position concurrent with or after commanding the first valve from the open position to the closed position.
9. The assembly of claim 6, wherein the instructions include instructions to: determine whether an occupant of the seat is above a size threshold; in response to determining the occupant of the seat is not above the size threshold, command the first valve from the open position to the closed position a first amount of time after commanding the first valve from the closed position to the open position; and in response to determining the occupant of the seat is above the size threshold, command the first valve from the open position to the closed position a second amount of time after commanding the first valve from the closed position to the open position, the second amount of time being greater than the first amount of time.
10. The assembly of claim 9, wherein the instructions include instructions to command the exhaust valve from the closed position to the open position concurrent with, or after, commanding the first valve from the open position to the closed position.
11. The assembly of claim 7, further comprising a second seat, and wherein the instructions include instructions to, in response to both determining the second seat is occupied and detecting the certain impact, command the second valve from a closed position to an open position.
12. The assembly of claim 11, wherein the instructions include instructions to command the second valve from the closed position to the open position a threshold amount of time after commanding

the first valve from the closed position to the open position.

13. The assembly of claim 12, wherein the threshold amount of time is 20-50 milliseconds.

14. The assembly of claim 11, wherein the instructions include instructions to command the first valve from the open position to the closed position after commanding the first valve from the closed position to the open position, and to command the second valve from the open position to the closed position after commanding the second valve from the closed position to the open position.

15. The assembly of claim 14, wherein the instructions include instructions to command the exhaust valve from a closed position to an open position concurrent with or after commanding the second valve from the open position to the closed position.

16. The assembly of claim 11, wherein the instructions include instructions to: determine whether an occupant of the seat is above a size threshold; in response to determining the occupant of the seat is not above the size threshold, command the first valve from the open position to the closed position a first amount of time after commanding the first valve from the closed position to the open position; in response to determining the occupant of the seat is above the size threshold, command the first valve from the open position to the closed position a second amount of time after commanding the first valve from the closed position to the open position, the second amount of time being greater than the first amount of time; and command the second valve from the open position to the closed position.

17. The assembly of claim 16, wherein the instructions include instructions to command the exhaust valve from the closed position to the open position concurrent with or after commanding the second valve from the open position to the closed position.

18. The assembly of claim 4, wherein the chamber defines a first cross-sectional area and the first fluid outlet defines a second cross-sectional area that is less than the first cross-sectional area.

19. The assembly of claim 4, wherein the manifold and the inflator are supported by the seat.
